

LLM 7-Days Report

Training tarayonida quyidagi
3ta namun narsa bor
- Test / - Anson / - Xatoliklar

Trainingda asosiy 3ta qism
ma'rufl

- ① Hyperparameters
- ② Training stability
- ③ Regularization

Training Hyperparameters

- ① Learning Rate (LR) - xatodan chiqariladigan
- ② Batch Size
- ③ Optimizer
- ④ Warm-up steps
- ⑤ Weight-decay

Training Stability

Kop weight ketadi

Kop pul ketadi

Stability juda muhim

LLM dagi Muammolar

Exploding Gradients - kop
lashlar, warmup qilinmaganda

Falkland Islands
(Islas Malvinas)
(administered by U.K.
claimed by ARGENTINA)

South Georgia and the
South Sandwich Islands
(administered by U.K.
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Coordinated Universal Time
formerly
Greenwich Mean Time

Sun
12:00
(number indicates standard time
in zone when it is 12 noon UTC)

juda ko'p ra xato matnlar
berilganda gradient partlashd

Muammo qanday bilinadi

① Loss kamayish o'rniga juda ko'p
olib ketadi

② Loss = Man yoki Loss = Inf

Muammoga yechim.

① LR warmup (shuncha bolan LRni
showish avval 0 dan boshlash h
masalan 2000 qadamgacha)

② Gradient Clipping (matematik
chegara, xar bor gradientni
threshold qiymat (threshold = 1)
dan oshib ketsa o'ziga qaytar)

Regularization

Regu bu overfittingni oldini
olish jarayond LLM da bu ham
boladi

weight Decay (training jarayonida
boron neyronning weight xaddan
tashqari katta boladi va uni
o'ziga qaytaradi)

Dropout (hamon bor qismi tashlab
yuboriladi)

LLM da faqat weight decay zararli
dropout = 0.0

LLM Architecture:

Model Building.

Components: Norm position, Norm type / Boos terms / MLP activation position embedding / Attention for inference / Stability - output stability - attention.

Attention Variants

MHA / GQA / MQA

Training stability interventions

Pre Norm, RMS Norm, Double-Norm, Z-loss, QK-Norm, Logit Soft-Cap, Gradient Clipping

Token

input ^② $\Rightarrow$ Embedding ^③ $\Rightarrow$ Tokenization ^④ $\Rightarrow$ Normalization ^⑤ $\Rightarrow$ Attention.

Hugging Face.

Models / Datasets / Spaces.

Tokenizer

tokenizer.json
tokenizer-config.json
vocab.json
merges.txt

Model Configuration
 $\rightarrow$ config.json

Neural Network

$\rightarrow$ model.safetensors

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